ElderShield: System Architecture

Predictive Digital Twin for Stroke and Fall Prevention in Geriatric Care

Team The_Glitch

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A. RUNTIME ARCHITECTURE

1 Data layer

- Kaggle Stroke Prediction dataset (5,110 rows)
- Simulated vitals stream: heart rate, systolic BP, SpO2
- Patient inputs: age, BP, glucose, BMI, smoking

Planned: ESP32 + MPU6050 + MAX30102 wearable

2 AI engine

- XGBoost with monotonic constraints
- Class-imbalance weighting
- SHAP TreeExplainer: top 3 reasons
- Bands: LOW, MEDIUM, HIGH

Planned: SisFall 1D-CNN fall model

3 FastAPI backend

- POST /predict
- POST /whatif
- GET /live, POST /live/reset
- POST /fall, GET /events
- CORS enabled, runs locally

4 Dashboard

- index.html: risk card, SHAP reasons, what-if table
- live.html: vitals tiles, BP graph, status banner
- Plain HTML and JavaScript

5 Caregiver alert

- Status: STABLE, WARNING, ALERT
- Fall event card with name, time and message
- Alert rules are simple demo rules

Planned: SMS / WhatsApp alerts

B. OFFLINE TRAINING PIPELINE

healthcare-dataset-stroke-data.csv
Raw Kaggle data

→

clean.py
Drop id, remove 'Other' gender, fill BMI with median, one-hot encode

→

clean_stroke.csv
5,109 rows, 16 columns, no missing values

→

train_stroke.py
80/20 stratified split, XGBoost, metrics

→

models/stroke_xgb.pkl
Saved model, loaded by risk_engine.py

→

explain_stroke.py
SHAP summary and patient plots

C. KEY RESULTS (held-out test set: 1,022 patients, 50 stroke cases)

0.817 AUROC	**0.22** PR-AUC (random 0.049)	**0.80** Recall at threshold 0.3	**0.105** Precision at threshold 0.3

Implemented in prototype Planned

Notes: live vitals and the fall event are simulated (no hardware connected yet). The risk score is a risk score, not a calibrated probability. What-if values are model-based estimates. ElderShield is decision support, not a diagnosis.